THE SPORADIC ACTIVITY OF PLIVA VALLIS, THE OUTLET VALLEY OF JEZERO'S CRATER LAKE, MARS.

J. Villette[1], N. Mangold[1], E. S. Kite[2], S. J. Conway[1] and L. Le Deit[1], [1]Nantes Université, Univ Angers, Le Mans Université, CNRS, Laboratoire de Planétologie et Géosciences, UMR 6112, Nantes, France (justine.villette@univ-nantes.fr), [2]University of Chicago, Chicago, IL, USA.

Introduction: Orbital and *in situ* observations have revealed that Jezero crater (the landing site for the Perseverance rover sent as part of the Mars 2020 mission) once hosted a paleolake [1-3]. Current observations suggest that Jezero crater was fed by two inlet valleys [1]. Sediments transported by these two valleys are at the origin of two depositional fans, interpreted as deltaic deposits [1, 2]. An outlet valley, called Pliva Vallis, is also present on the eastern crater rim [1] and its presence raises the question of whether the lake system operated mainly as an open basin, or as a closed basin system with episodic overflows. Resolving this uncertainty would improve our understanding of the evolution of Pliva Vallis with time and consequently our knowledge of the history of Jezero's lake, a site of interest for understanding ancient martian hydrologic activity, and for finding potential traces of past life.

Observations and interpretations: We conducted a detailed morphological study of Pliva Vallis using digital elevation models and orbital images (CTX and HiRISE data) acquired by Mars Reconnaissance Orbiter [4-6].

Rock exposures in the valley. HiRISE images reveal stratified sedimentary deposits on the floor and inner walls of Pliva Vallis and suggest a fluvial activity in this valley. Alternating fine deposits and strata containing meter-scale boulders reflect the variation of flow intensity over time. The re-incision of some of these deposits suggests that they were re-incised by a flow event subsequent to the one responsible for their deposition.

Morphometry. We observed several topographical features that are atypical for overland flows [7]. These observations include a decrease of the valley width and depth from upstream to downstream, which is not consistent with a steady flow over time, the presence of a perched valley in the downstream part of the valley, and the presence of three main topographic rises on the valley floor which all together suggest a discontinuous and energetic fluvial regime of the outlet valley. These anomalies reveal the high competence of the bedrock and the difficulty of incision. HiRISE images and DTMs also reveal the presence of four topographical steps within the wall of the valley, which are marked by the four colored lines in Figure 1 and interpreted as four bedrock incisions terraces. The morphology of Pliva Vallis is more compatible with rapid flow events caused by several phases of breach than a continuous outlet activity.

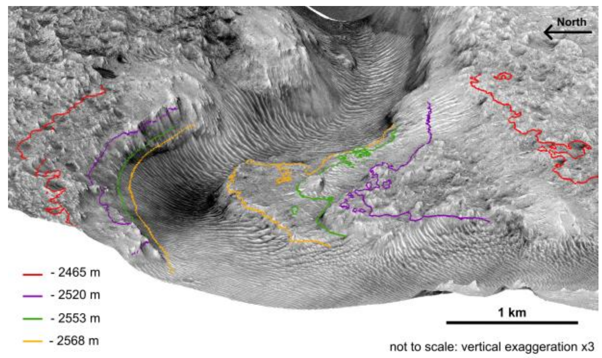

Fig. 1: vertically exaggerated 3D representation (HiRISE image over a DTM) of Pliva Vallis where the four incision terraces were identified. Each colored line identifies the elevation of these terraces.

Scenario: Our observations lead us to interpret the formation of Jezero outlet valley by discontinuous and episodic flows, responsible for the progressive and sporadic incision of the valley. Thus, we propose that the lake overflowed sporadically during the four breach events before emptying completely as a closed system though evaporation and infiltration. Hereafter, we explore this hypothesis using numerical modeling.

Modeling of the breach erosion processes: To give a first estimate of the duration of these four episodes of lake overflow and valley incision, we use a 0-D model, simulating valley formation by breach erosion processes and crater lake spill over [8, 9]. The model requires us to make some simplifying assumptions (e.g., uniform valley geometry, no pre-existing valley, no downstream change in flow velocity, etc.), but provides a simple and rapid simulation of breach formation, particle transport and erosion rate in the outlet valley over time. We apply the model four times to Jezero's crater lake in order to simulate the four overflow episodes previously identified.

For each episode, we consider the valley geometry, such as the valley width, measured for each episode in Figure 1, and the slope, measured at the breach. The slope of the first episode corresponds to the slope of the outer crater rim (0.075) and for the latest the slope used is the slope of the valley floor measured at the breach (0.022). For the second and the third episodes we tested a range of slope between the slope of the fourth episode (0.022) and the slope of the first episode (0.075). According to the initial flow depth, which is the flow depth at the breach, we tested a range of realistic values for each episode. Lastly, we assume an incoming flow into the crater of about 100 $m^3.s^{-1}$ from previous studies on Neretva Vallis [10], and we found that varying this value has little effect on our results given the order of magnitudes differences in discharge rate between this steady-state process and the overflows.

Figure 2A shows how particle transport (D50 of 0.076 m [10]) changes over time for the example of the second overflow event, for a range of slopes and a range of initial flow depth. Modeling results suggest that the transport stage decreases over time, which is due to the progressive fall in discharge after breaching. As soon as the breach occurs, the flow loses energy progressively at first and then rapidly. When the transport stage reaches a value of 1, the flow is too weak to continue to carry particles.

The erosion rate can also be calculated and is shown in Figure 2B for the second overflow event. The erosion rate follows the same trend as the transport stage over time: it decreases slowly at first, then rapidly when the discharge is no longer sufficient to maintain the erosion of the valley.

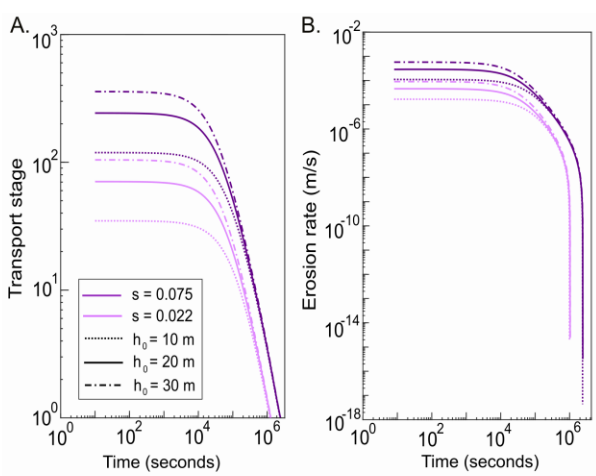

Fig. 2: Modeling results for the second episode. A: Transport stage as a function of time; B: Erosion rate ($m.s^{-1}$) as a function of time. We tested two slopes (s=0.075 in dark purple and s=0.022 in light purple) and a range of initial flow depth (h_0).

We present here only the results of the second episode of incision and overflow. However, each episode has been simulated and the results obtained for particle transport and erosion follow the same trend as for the second episode.

In terms of timescale, modeling results suggest that particle transport and valley erosion lasted from few days to few weeks per event. The timescales calculated with this model are consistent with our interpretation that the flow events in the outlet valley were discontinuous and short term.

While exploring the sensitivity of the model to the parameter space we found that the initial flow depth does not have a big effect on the duration of sediment transport and erosion. However, it influences the efficiency of sediment transport and therefore of erosion, particularly at the beginning of the lake overflow. Lastly, the slope affects the duration of sediment transport and erosion in the valley. However, the time scale is little-changed and remains very rapid.

Conclusion: The study of the outlet valley of Jezero crater helps us to better understand the activity of the associated lake. Our observations allow us to propose a new scenario for the evolution of Pliva Vallis and consequently for the evolution of the Jezero's crater lake - a closed basin system which was sporadically opened during at least four identified overflow events.

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